Windows Updates for Business

Implementation Solution Guide

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Executive Summary — Windows Updates for Business (Microsoft Intune)

White-label · Intune Solutions

Unpatched devices are how breaches start, and uncontrolled updates are how productivity dies — surprise reboots, version drift, and a fleet nobody can report on. **Windows Update for Business**, managed through Microsoft Intune, replaces WSUS and guesswork with **update rings**: a phased, scheduled, policy-driven rollout of quality and feature updates, the ability to **expedite** critical zero-day fixes, and clear **compliance reporting** across every device.

This is a complete, **deploy-it-today** package for implementing Windows Update for Business with Microsoft Intune, written for an administrator with no prior update-management experience.

What you get

- A plain-language **Solution Overview** of rings, feature updates, expedited updates, and Delivery Optimization.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for update-status discovery, ring validation, fleet compliance reporting, safe remediation, and ring/assignment auditing.
- **Testing** and **Troubleshooting** guides for confident rollout.

What this delivers

A **ring-based update strategy** — a fast **Pilot ring** and a slower **Broad ring** — with quality and feature-update deferrals, deadlines and active hours, a **feature update profile** that pins the target Windows version, an **expedite** path for emergencies, and Delivery Optimization tuned to spare your bandwidth.

The standard deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Patched, predictable, reportable devices — no surprise reboots, no version drift, and a zero-day response button — deployable by a single administrator and validated before fleet rollout.

1. Solution Overview — Windows Updates for Business

What Windows Update for Business is

Windows Update for Business (WUfB) is Microsoft's cloud service for controlling how and when Windows devices receive **quality updates** (monthly security/reliability patches) and **feature updates** (annual OS versions). Managed through Microsoft Intune, it replaces WSUS for cloud-managed devices: you define **update rings** that defer, schedule, and deadline updates, and the devices pull updates directly from Microsoft with **Delivery Optimization** peer caching.

How it works

1. An Intune **Update ring** sets quality- and feature-update **deferral** periods, **deadlines**, **active hours**, restart behavior, and the user experience.
1. Rings are assigned to groups in a **ring strategy** — a small **Pilot** ring updates first (short deferral), a **Broad** ring follows (longer deferral) once the pilot is healthy.
1. A **Feature update profile** pins the target OS version (e.g. Windows 11 23H2) so devices don't drift ahead or fall behind; you control the move to the next version.
1. **Expedited updates** push a specific critical/zero-day patch immediately, bypassing deferrals.
2. **Delivery Optimization** lets devices share update content on the LAN to save WAN bandwidth.
3. Update status is reported in Intune (and, optionally, Windows Update for Business reports).

What you can control

- **Quality update deferral** (0–30 days) and **deadline** with grace period.
- **Feature update deferral** / **target version** (pin the OS build).
- **Active hours** and **auto-restart** behavior; user notifications and the option to pause.
- **Expedite** a named quality update for emergencies.
- **Driver updates** (approve/defer) and **Delivery Optimization** download mode/bandwidth.

Benefits

- **Security** — devices stay patched on a predictable cadence; zero-days can be expedited.
- **Stability** — ring-based rollout catches a bad update on the pilot before it hits the fleet.
- **No version drift** — feature update profiles keep the fleet on a known, supported build.
- **Bandwidth-friendly** — Delivery Optimization reduces WAN load.
- **Visibility** — per-device and fleet-wide update status and feature-version inventory.

Limitations (be honest with buyers)

- WUfB manages **Microsoft** updates (Windows + drivers + Microsoft Store); third-party app patching is separate.
- Rich Windows Update for Business **reports** need diagnostic data enabled and a Log Analytics workspace (optional add-on, noted in requirements).

• Feature updates are large; first rollout timing should account for bandwidth.

• Co-management with ConfigMgr/WSUS requires deciding which authority owns updates (avoid conflicts).

The standard deployment model

> ***Create Pilot + Broad update rings*** → ***add a feature update profile*** → ***assign by ring*** → ***validate on the pilot*** → ***expand to broad.*** *repeatable pattern:* ***Assess*** → ***Configure*** → ***Pilot*** → ***Validate*** → ***Expand.***

2. Requirements Guide — Windows Updates for Business

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1). WUfB itself is a free Windows feature. **Why:** Intune delivers the update rings/profiles and reports status. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

2.2 Device enrollment

Description: Targeted devices enrolled in Intune. **Why:** Update rings apply only to enrolled devices. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.3 Supported Windows OS

Description: Windows 10/11 (Pro/Enterprise/Education). **Why:** WUfB policies (rings, feature profiles, expedite) require a supported build. **Verify:** `winver`; run `WUFB-Discovery.ps1`. **Implement:** Patch/upgrade to a supported build.

2.4 Network (Windows Update + Delivery Optimization)

Description: Outbound HTTPS to Windows Update and Delivery Optimization endpoints. **Why:** Devices pull updates from Microsoft and share content via Delivery Optimization. **Verify:** From a device, Windows Update can check online; DO endpoints reachable. **Implement:** Allow Windows Update / Delivery Optimization URLs through proxy/firewall; do not block DO peer ports if you want LAN caching.

2.5 Single update authority (no conflicts)

Description: Intune/WUfB is the authority for updates — not a leftover WSUS/GPO or ConfigMgr. **Why:** Two update authorities fight and stall updates. **Verify:** Check for a `WUServer` GPO/registry value; run `WUFB-Discovery.ps1` (flags WSUS). **Implement:** Remove the WSUS/GPO settings (or set co-management update workload to Intune).

2.6 (Optional) Windows Update for Business reports

Description: Diagnostic data enabled + a Log Analytics workspace for the WUfB reports. **Why:** Required only for the richer Update Compliance / WUfB reporting experience. **Verify:** Devices send **Required** diagnostic data; the workspace is configured. **Implement:** Set diagnostic data via an Intune policy and connect Windows Update for Business reports (optional).

Checklist

- Intune licensing assigned
- Devices enrolled
- Windows 10/11 supported build — `WUFB-Discovery.ps1`
- Network allows Windows Update + Delivery Optimization
- No conflicting WSUS/GPO/ConfigMgr update authority
- (Optional) Diagnostic data + Log Analytics for WUfB reports
- Pilot + Broad Entra groups created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Windows Updates for Business

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *A fast **Pilot** ring and a slower **Broad** ring, plus a feature update profile that pins the target Windows version. Pilot validates every update before the fleet gets it.*

Step 1 — Create the ring groups

1. Entra → **Groups** → **New group** → Security → `WUfB-Pilot` (a few representative devices). **Create**.
2. Create `WUfB-Broad` (the rest of the fleet). **Create**.

Step 2 — Create the Pilot update ring

1. Intune → **Devices** → **Windows** → **Update rings for Windows 10 and later** → **Create profile**.
2. **Name:** `Update Ring - Pilot`. **Next**.
3. **Update settings:**

• **Quality update deferral period (days): 0**.

- **Feature update deferral period (days): 0**.
- **Automatic update behavior:** Auto install and restart at a scheduled time (or at maintenance time).
- **Active hours:** 8:00–18:00 (adjust to your org).
- **Deadline for quality updates: 2 days**; **feature updates: 3 days**; **grace period: 2 days**.
- **User experience:** allow the user to pause = your policy.

1. **Next** → **Assignments** → **Add groups** → `WUfB-Pilot`. **Next** → **Create**.

Step 3 — Create the Broad update ring

1. **Create profile** again. **Name:** `Update Ring - Broad`.
2. **Update settings: Quality deferral 7 days**, **Feature deferral 14 days**, deadlines as above.
3. **Assignments** → **Add groups** → `WUfB-Broad`. **Create**.

Step 4 — Create the feature update profile (pin the version)

1. Intune → **Devices** → **Windows** → **Feature updates for Windows 10 and later** → **Create profile**.
2. **Name:** `Feature Update - Target`. **Feature update version:** select your target (e.g. **Windows 11, version 23H2**).
3. **Rollout options:** Make update available as soon as possible (or schedule a gradual rollout).
4. **Assignments** → **Add groups** → `WUfB-Pilot` and `WUfB-Broad`. **Create**.

Step 5 — Configure Delivery Optimization (optional but recommended)

1. **Devices** → **Configuration** → **Create** → **Settings catalog** → search **Delivery Optimization**.
2. Set **Download Mode** to **HTTP blended with peering behind same NAT (1)** or **LAN (1)**; set max bandwidth as needed. Assign to all update groups. **Create**.

Step 6 — Apply on a pilot device

1. On a pilot device, sync policy (**Settings** → **Accounts** → **Access work or school** → **Info** → **Sync**).
2. Confirm the deferral/deadline settings applied (**Settings** → **Windows Update** → **… managed by your organization**), or run `WUFB-Validation.ps1`.

Step 7 — Validate the pilot

1. Run `WUFB-Validation.ps1` → confirms the ring policy applied (deferrals/deadlines in registry).
2. Run `WUFB-ComplianceReport.ps1` → confirms pilot devices' update status and OS build.
3. Let the next quality update install on the pilot and confirm a healthy result. Complete the **Testing Guide**.

Step 8 — Expedite path (for emergencies)

1. When a zero-day quality update is released, go to **Devices** → **Windows** → **Quality updates (Expedite)** → **Create profile**, choose the update, set restart grace, and assign — it bypasses deferrals.

Step 9 — Expand to production

1. The **Broad** ring already covers the fleet; confirm it's healthy after the pilot passes each cycle.
2. Move the feature update target forward only after the pilot validates the new version.
3. Re-validate each cycle with `WUFB-ComplianceReport.ps1` / `WUFB-RingReport.ps1`.

Order of operations

`Ring groups → Pilot ring → Broad ring → Feature update profile → Delivery Optimization → Apply → Validate → Expedite (as needed) → Expand`

5. Testing Guide — Windows Updates for Business

Run after the pilot ring applies (Implementation Step 7) and after each update cycle / feature move.

Test 1 — Ring policy applied

Do: Intune → **Devices** → **Windows** → **Update rings** → `Update Ring - Pilot` → **Device status**. **Expected:** Pilot devices show **Succeeded**.

Test 2 — Settings present on device

Do: On a pilot device run `WUFB-Validation.ps1`, or open **Settings** → **Windows Update** ("Some settings are managed by your organization"). **Expected:** Quality/feature deferral and deadline values match the ring; active hours set.

Test 3 — Updates install on schedule

Do: Let the current quality update flow to the pilot; observe install + scheduled restart within the deadline. **Expected:** The update installs and the device restarts per the deadline/active-hours policy.

Test 4 — Feature version pinned

Do: Run `WUFB-ComplianceReport.ps1` (or check `winver`). **Expected:** Pilot devices are on (or moving to) the **targeted** feature version — not drifting ahead.

Test 5 — Expedite works (when needed)

Do: Create an expedite profile for a test/recent quality update; assign to a pilot device. **Expected:** The device installs that update promptly, bypassing the deferral.

Test 6 — Delivery Optimization active

Do: On a pilot device: `Get-DeliveryOptimizationStatus` after an update downloads. **Expected:** Peer/LAN bytes are used where available (download mode reflects the policy).

Test 7 — Fleet report

Do: Run `WUFB-ComplianceReport.ps1` and `WUFB-RingReport.ps1`. **Expected:** A clear per-device update status + OS build inventory and the ring assignments.

Pass/fail gate

Gate	Requirement
Pilot pass	Tests 1–4 pass; the cycle's quality update installs cleanly on the pilot
Broad pass	Tests 1–3, 7 pass for the broad ring before the next feature move

6. Troubleshooting Guide — Windows Updates for Business

Use `WUFB-Validation.ps1` / `WUFB-Discovery.ps1` for device state and `WUFB-ComplianceReport.ps1` for fleet status, `WUFB-Remediation.ps1` for safe fixes.

A. Updates not installing / stuck

Symptom	Cause	Remediation
Device never updates	Not synced, deferral too long, or paused	Run `WUFB-Remediation.ps1` (sync + USO scan); check deferral/deadline; ensure not paused
"Some settings managed" but no updates	Conflicting WSUS/GPO authority	Run `WUFB-Discovery.ps1` (flags `WUServer`); remove WSUS/GPO; for co-management set update workload to Intune
Ring shows Error	Device not enrolled/checked in	Confirm enrollment and group membership; sync

B. Surprise / unwanted restarts

- Tune **active hours** and the **auto-restart / deadline grace** in the ring; communicate the deadline to users.
- For sensitive groups, lengthen the grace period or allow user-scheduled restart.

C. Feature update issues

Symptom	Cause	Remediation
Device jumped to an unexpected version	No feature update profile pinning the version	Create/assign a Feature update profile with the target version
Feature update won't offer	Safeguard hold, or device below target	Check compatibility/safeguard holds; confirm the target version and rollout setting

D. Expedite not working

• Expedite requires the Windows Update service healthy and a supported build; confirm the device isn't paused and the Update Health Tools are present (they install automatically on supported builds).

E. Bandwidth / Delivery Optimization

• WAN saturated on feature-update day → set Delivery Optimization download mode to LAN/peer and cap foreground/background bandwidth; stage feature rollout gradually.

F. Diagnostics quick reference

```
Get-ComputerInfo | Select WindowsProductName, OsBuildNumber, WindowsVersion
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate' -EA SilentlyContinue   # deferr
usoclient StartScan        # trigger an update scan
Get-DeliveryOptimizationStatus    # DO peer/LAN usage
```

Update logs: `Get-WindowsUpdateLog` (generates a merged log on the desktop).

G. When to escalate

If the ring shows Succeeded, no WSUS/GPO conflict exists, the device is synced and not paused, and updates still won't install after a USO scan, collect `Get-WindowsUpdateLog` output and open a Microsoft support case.

7. Best Practices — Windows Updates for Business

• **Use rings, always.** A small fast **Pilot** ring catches a bad update before it reaches the **Broad** fleet — the single most important practice.

• **Pin the feature version.** A Feature update profile prevents drift and keeps the fleet on a known, supported build; move it forward only after the pilot validates the new version.

• **Set realistic deadlines + active hours.** Patches must install on a schedule, but respect working hours and give a grace period to avoid surprise reboots.

• **One update authority.** Let Intune/WUfB own updates — remove leftover WSUS/GPO settings and, under co-management, set the update workload to Intune.

• **Keep an expedite path ready.** Pre-plan the expedite profile so a zero-day can ship in minutes, bypassing deferrals.

• **Tune Delivery Optimization.** Use LAN/peer caching and bandwidth caps so feature-update day doesn't saturate the WAN; stage large rollouts.

• **Communicate to users.** A short note about the deadline and restart behavior dramatically reduces tickets.

• **Watch the reports.** Track update status and OS-build inventory with `WUFB-ComplianceReport.ps1` (or Windows Update for Business reports) and chase laggards.

• **Don't over-defer.** Long deferrals leave devices exposed; keep quality-update deferrals short (0 pilot / ~7 broad).

• **Document the cadence.** Record ring deferrals, deadlines, and the target feature version for audits and repeatability.

8. Deliverables Summary — Windows Updates for Business

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What WUfB is, rings/feature/expedite/Delivery Optimization, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered Intune steps (rings, feature profile, expedite, DO)
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`WUFB-Discovery.ps1`	Reports OS build, update deferral settings, and any conflicting WSUS authority
`WUFB-Validation.ps1`	Confirms the ring policy applied (deferrals/deadlines present)
`WUFB-ComplianceReport.ps1`	Fleet-wide update status + OS-build inventory (Microsoft Graph)
`WUFB-Remediation.ps1`	Safe remediation; forces a sync and a Windows Update scan (USO)

Script	What it does
`WUFB-RingReport.ps1`	Lists Intune update rings + feature profiles and their assignments (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Patched, predictable, reportable devices — phased ring rollout, pinned feature version, and a zero-day expedite button.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WUFB-ComplianceReport.ps1

WUFB-Discovery.ps1

WUFB-Remediation.ps1

WUFB-RingReport.ps1

WUFB-Validation.ps1